RESEARCH PROPOSAL REPORT

Longitudinal Multimodal Progression Modelling of Osteoradionecrosis of the Jaw

Fusing Serial Imaging, Salivary Biomarkers, and Clinical Data for Personalized Staging and Treatment Triage

PREPARED FOR
Prof. Vaishali Kulkarni - Design Experience Evaluation

INSTITUTION
NMIMS MPSTME, Mumbai

AUTHORS
Arnav Jain · Seher Siddiqui

DATE
April 2026

1. Broad Domain of Work

Primary Domain: Head & Neck Radiation Oncology ⟶ Post-Treatment Complication Monitoring

Sub-domain: Temporal Multimodal Machine Learning for Longitudinal Disease Progression Modelling

This research sits at the convergence of three fields: radiation oncology (specifically late toxicities of head and neck radiotherapy), medical imaging and radiomics (CT, CBCT, OPG feature extraction), and multimodal machine learning (temporal fusion architectures, survival modelling, and clinical decision support). The immediate clinical context is osteoradionecrosis of the jaw (ORNJ) at Tata Memorial Hospital — Asia's largest cancer centre and the only high-volume oncology institution in India with the case mix, imaging archive, and multidisciplinary expertise to support this work.

ONE-LINE SUMMARY

We are building a model that does not merely predict **who will get ORN —** but one that tracks **how it progresses over time** and tells you precisely **what to do about it.**

2. Introduction

2.1 Disease Overview: What is Osteoradionecrosis?

Osteoradionecrosis (ORN) is a severe, progressive late complication of radiation therapy (RT) for head and neck cancers (HNC), defined as non-healing, exposed bone within the irradiated field in the absence of tumour recurrence. The underlying pathophysiology follows Marx's triad: hypoxia, hypovascularity, and hypocellularity of irradiated tissue — leading to impaired wound healing, progressive bone death, and ultimately, structural failure of the jaw.

Head and neck cancers represent the **seventh most prevalent malignancy globally**, affecting over **890,000 individuals annually** — a number rising due to HPV-associated oropharyngeal cancer in younger populations. As survival rates improve (particularly in HPV-positive patients), the burden of late radiation toxicities is growing. Mandibular ORN affects an estimated **3–15%** of all HNC patients treated with RT, with incidence higher in those receiving concurrent chemotherapy, dental extractions within the radiation field, or doses exceeding 60 Gy.

2.2 Why ORN Is Uniquely Difficult to Manage

Once ORN develops, clinicians face a decision tree of escalating treatment intensity: conservative antibiotics and debridement → PENTOCLO protocol (pentoxifylline + tocopherol + clodronate) → hyperbaric oxygen (HBO) therapy → major resective surgery with free flap mandibular reconstruction. The choice of pathway depends entirely on the severity and trajectory of the disease — but current staging is subjective, inconsistent across specialties, and takes no account of how the disease is evolving over time.

A landmark 2025 interobserver study at MD Anderson involving 20 clinicians across oral oncology, radiation oncology, surgery, and neuroradiology demonstrated **no specialty showed superior diagnostic accuracy** in ORN staging — with significant inter-specialist variability documented. HBO therapy, frequently prescribed, shows only a **~38% success rate when used in isolation**, yet there is currently no validated model to predict which patients will actually respond to it.

2.3 The Critical Gap This Research Addresses

The entire ML literature on ORN to date treats it as a binary, single-timepoint problem: will this patient develop ORN (yes/no), computed from pre-treatment data. This ignores the biological reality that ORN is a dynamic, progressive disease that evolves across months to years post-radiation.

THE GAP IN ONE PARAGRAPH

No existing model: (1) tracks ORN disease state across multiple timepoints; (2) integrates imaging + biomarkers + clinical variables simultaneously; (3) outputs a treatment pathway recommendation; or (4) uses an Indian/South Asian patient cohort. This research proposes all four simultaneously — making it genuinely novel in the global literature.

Additionally, the CT-based ORN grading system proposed by **Patil et al. (2026)** from Tata Memorial Hospital itself — published in *Cancers* — provides a locally validated radiological taxonomy upon which a longitudinal ML model can be directly built, giving this research an institutional lineage and clinical anchor that no international group can replicate.

Patil V et al. Cancers 2026;18:187 — CT-based grading system from Tata Memorial (DOI: 10.3390/cancers18020187)

2.4 Why This Approach and Why Now

This research specifically pursues **Longitudinal Progression Modelling + Treatment Triage** — a strategy that is harder to execute but vastly more impactful than binary risk prediction. The case for this angle rests on four pillars:

- **Clinical urgency:** The question that actually confronts the treating clinician is not 'will this patient get ORN?' (answered months ago) but 'my patient has Grade 2 ORN — will it progress, and should I operate now or wait?' No validated tool addresses this.
- **Scientific gap:** The Barua et al. study is the only published longitudinal imaging paper in ORN; it involved 21 patients, no biomarkers, and no ML. The West et al. (2026) dataset is longitudinal but contains no imaging. The field is wide open.
- **Institutional advantage:** Tata Memorial's Patil et al. (2026) CT grading taxonomy, combined with its retrospective archive of 300+ HNC patients per year since 2015, creates a unique data asset. The South Asian patient demographic (bidi/smokeless tobacco, betel nut, high-grade oral cavity primaries) is independently publishable as a novel cohort.
- **Journal positioning:** A validated longitudinal multimodal ML model with treatment triage outputs is competitive for Lancet Digital Health (IF 23.8) or Nature Medicine — tiers unreachable with a binary risk predictor.

3. Approaches Explored in the Literature

The following paradigms represent the complete landscape of ML approaches applied to ORN, drawn from the reference papers in this study's Drive folder. All share a fundamental limitation: they are static, single-timepoint, and focus exclusively on binary risk prediction.

3.1 DVH-Based Classical Machine Learning (Humbert-Vidan et al., 2021)

The foundational ML study in ORN prediction. Using a 96-patient case-control cohort from Guy's & St Thomas' Hospital (London), Humbert-Vidan et al. compared logistic regression (LR), SVM, random forest (RF), AdaBoost, and artificial neural networks (ANN) using **dose-volume histogram (DVH) parameters** as the sole input features. The ANN achieved 77% accuracy, followed closely by SVM (76%) and AdaBoost (75%). Crucially, McNemar's test found **no statistically significant difference between any of the models** — indicating the input features (DVH parameters) are the binding limitation, not the model architecture.

Humbert-Vidan L et al., Br J Radiol. 2021;94:20200026 (DOI: 10.1259/bjr.20200026)

3.2 Deep Learning on 3D Dose Maps vs Classical ML (Reber et al., 2023)

The largest direct comparison in the ORN ML literature (MD Anderson; n = 1,236–1,259 patients). Compared logistic regression, RF, SVM, and ResNet/DenseNet/autoencoder-based CNNs using **full 3D dose distribution maps** cropped to the mandible as DL inputs versus DVH summary statistics for classical models. Logistic regression (F1 = 0.3) outperformed all DL models — with ResNet achieving F1 = 0.07 and DenseNet F1 = 0.14. Increasing DL training data did not improve performance. This critically demonstrates that **spatial dose information alone is insufficient** and that additional imaging modalities and non-dosimetric data must be incorporated.

Reber B et al., Adv Radiat Oncol. 2023;8(4):101163 (DOI: 10.1016/j.adro.2022.101163)

3.3 Multi-Institutional NTCP Prediction: PREDMORN Study (Humbert-Vidan et al., 2025)

The most rigorous multi-centre study in the field. Across 8 institutions (n = 3,928 patients; 622 ORN cases), a forward stepwise logistic regression NTCP model was developed and externally validated using **clinical, demographic, and dosimetric variables**. Key predictors included mandibular D30%, dental extractions, post-operative RT, and smoking. The model demonstrated good discrimination but remained a static, pre-treatment binary classifier. Its significance lies in demonstrating that **DVH-based NTCP models do generalize** across institutions when clinical covariates are included — a baseline against which any temporal model must be benchmarked.

Humbert-Vidan L et al., Int J Radiat Oncol Biol Phys. 2025 — PREDMORN multi-institutional study (PMC)

3.4 Unsupervised ML: Cluster-Based DVH Risk Stratification (Hosseinian et al., 2023)

A methodologically novel approach using **K-means clustering on whole DVH curves** (rather than summary statistics) to identify six distinct mandibular dose-volume risk regions, followed by soft-margin SVM boundary fitting. Applied to 1,259 MD Anderson patients, this enriched dose-volume risk stratification beyond single-parameter NTCP models. While not a prediction model in the classical sense, it showed that **unsupervised structural analysis of dose data** reveals clinically meaningful risk strata. This approach will be extended to cluster longitudinal imaging trajectories in the proposed model.

Hosseinian S et al., medRxiv 2023 (DOI: 10.1101/2023.03.24.23287710)

3.5 Salivary Cytokines as Predictive Biomarkers (Principe et al., 2019 & 2022)

Two studies from Valencia examining salivary cytokine panels (IL-1β, IL-6, IL-8, IL-10, MCP-1, TNF-α, VEGF, EGF) in HNC patients undergoing RT. The 2022 study found that **low pre-RT salivary IL-8 independently predicts positive RT response** (AUC = 0.84). The 2019 review confirmed RT-induced increases in IL-1β, IL-6, IL-8, MCP-1, and TNF-α correlate with irradiation dose. These findings establish the biomarker input branch for the proposed model — **no ORN-specific ML paper has yet integrated salivary cytokines** as temporal input features alongside imaging.

Principe S et al., J Clin Exp Dent. 2019;11(11):e1072-7 — Salivary cytokines review

Principe S et al., Clin Oral Investig. 2022;26:437–448 — IL-8 as RT biomarker (DOI: 10.1007/s00784-021-04017-0)bvb

3.6 Tata Memorial CT Grading System (Patil et al., 2026)

A retrospective cohort of 35 ORN patients from Tata Memorial Hospital, examining CT imaging characteristics at diagnosis and follow-up. The study proposes a **novel CT-based grading system (Grades 0–4)** with intraosseous gas and fistula modifiers (G+/G−, F+/F−) based on purely radiomorphological criteria. This is a direct precursor to the proposed ML model — it provides the imaging taxonomy, documents the progression patterns (increased erosion, fragmentation, and sclerosis on follow-up), and originates from the same institution.

Patil V et al., Cancers 2026;18:187 — Tata Memorial CT grading (DOI: 10.3390/cancers18020187)

3.7 Longitudinal Dataset for ORNJ NTCP Modeling (West et al., 2026)

A data descriptor paper from MD Anderson presenting a **longitudinal cohort of 1,129 HNC patients** with structured follow-up at 3–6, 12, 18, and 24 months and annually post-RT. This dataset underpins a Weibull Accelerated Failure Time (WAFT) survival model for ORNJ timing prediction. Its existence validates the proposed data architecture and serves as a benchmark comparison cohort — the closest existing dataset to what is proposed at Tata Memorial.

West NA et al., Sci Data. 2026;13:16 — Longitudinal ORNJ dataset (DOI: 10.1038/s41597-025-06321-w)

3.8 The Only Longitudinal Imaging Study: Barua et al.

A 21-patient retrospective series examining serial CT characteristics in ORN over follow-up periods of 6–36 months. This represents the **only published paper tracking imaging changes longitudinally in ORN** — yet it contains no ML, no biomarkers, and no treatment outcome analysis. Its small size and purely descriptive methodology define the floor that the proposed model must surpass.

SUMMARY: WHAT THE LITERATURE HAS NOT DONE

- **No temporal ML model:** All models predict a static binary outcome from a single pre-treatment snapshot
- **No biomarker integration:** No ORN ML paper has combined imaging with salivary/serum biomarker data
- **No treatment triage output:** No model tells you which treatment a patient should receive based on current disease trajectory
- **No Indian cohort:** Every large dataset is from MD Anderson (Texas) or European institutions — zero representation of South Asian populations with distinct risk profiles (bidi, smokeless tobacco, betel nut, oral cavity primaries)
- **No competing-risk survival modeling:** No study models time-to-progression vs. time-to-death as simultaneous competing events

4. Datasets Used by Reference Studies

All major ORN ML datasets share three characteristics: retrospective design, Western-institution origin, and single-timepoint data collection. The table below summarises the key datasets across the reference papers, followed by the proposed Tata dataset profile.

Study / Source	N (ORN+)	Modality	Timepoints	Key Limitation
Humbert-Vidan 2021 (KCL / Guy's London)	96 (48)	DVH + clinical	Single (pre-RT)	Tiny cohort; no imaging
Reber et al. 2023 (MD Anderson)	1,259 (173)	3D CT dose maps	Single (pre-RT)	DL underperformed; no biomarkers
PREDMORN 2025 (8 institutions)	3,928 (622)	DVH + clinical	Single (pre-RT)	No imaging; no temporal data
Hosseinian 2023 (MD Anderson)	1,259 (n/a)	DVH curves only	Single (pre-RT)	Unsupervised only; no prediction
West et al. 2026 (MD Anderson)	1,129 (followed)	Clinical + dosimetric	Multi (3–24mo)	No imaging; no biomarkers
Patil et al. 2026 (Tata Memorial)	35 (35)	CT + follow-up	Multi (initial + f/u)	Small n; no ML; no biomarkers
Barua et al. (case series)	21 (21)	Serial CT	Multi (6–36mo)	Purely descriptive; no ML
Principe et al. 2019 & 2022 (Valencia)	30 (HNC)	Salivary cytokines	Pre/post-RT	Not ORN-specific; small cohort

Study / Source	N (ORN+)	Modality	Timepoints	Key Limitation
PROPOSED: Tata Memorial Retrospective Cohort — Est. 80–150 ORN-confirmed HNC patients (2015–2024); serial CT/OPG at 3, 6, 12, 24 months; salivary/serum cytokine panel from follow-up biobank (if available); full clinical + dosimetric records. Graded using Patil et al. 2026 CT system. South Asian demographics (bidi, smokeless tobacco, betel nut, oral cavity primaries) = independent global novelty.				

WHY THE TATA DATASET IS GLOBALLY UNIQUE

- **South Asian risk factors:** Bidi smoking, smokeless tobacco (gutka/khaini), and betel nut chewing are prevalent in India and mechanistically distinct from cigarette smoking in terms of carcinogen exposure, oral mucosal injury, and RT response. No existing ORN ML dataset captures these.
- **Oral cavity primaries dominate:** Unlike Western datasets (predominantly oropharyngeal HPV+ cancers), Tata's case mix includes high rates of buccal mucosa, tongue, and floor-of-mouth primaries — anatomically closer to the mandible, with different dose geometry and ORN risk distribution.
- **Longitudinal follow-up infrastructure:** Tata's structured follow-up protocol means serial CT imaging is already embedded in standard care, creating a de facto longitudinal imaging archive that most institutions lack.

5. System Design: Flowchart, ML Architecture & Pseudocode

5.1 End-to-End System Overview

The proposed system has four stages: Data Ingestion & Preprocessing, Multimodal Feature Extraction, Temporal Fusion Modeling, and Clinical Output Generation. The flowchart below describes the full pipeline.

STAGE 1 — DATA INGESTION & PREPROCESSING

Inputs (4 parallel branches):

- **Branch A — Imaging:** Serial CT scans at T0 (pre-RT), T1 (3mo post-RT), T2 (6mo), T3 (12mo), T4 (24mo). DICOM preprocessing: resampling to 1×1×1 mm, HU windowing, mandible auto-segmentation using pretrained nnU-Net.
- **Branch B — Dosimetric:** 3D dose distribution maps from RT planning system (DICOM-RT). DVH parameter extraction: Dmax, D30%, D50%, mean dose, and dose to mandibular sub-volumes (alveolar vs. basilar bone).
- **Branch C — Salivary/Serum Biomarkers:** IL-6, IL-8, TNF-α, MCP-1 measured at each follow-up visit (T1–T4) via Luminex xMAP multiplex immunoassay. Serum CTX (bone resorption marker) and P1NP (bone formation marker) if available.
- **Branch D — Clinical Variables:** Baseline: dental health score, tobacco type (cigarette / bidi / smokeless / betel nut), diabetes, alcohol, concurrent chemotherapy, pre-RT extractions, HNC primary site, T-stage, N-stage, IMRT dose schedule.

▼

STAGE 2 — MULTIMODAL FEATURE EXTRACTION

- **CT Radiomics (PyRadiomics):** Extract 1,645 features per timepoint per mandibular sub-volume — first-order statistics, GLCM texture, GLRLM, GLSZM, GLDM, shape features. Apply MFPCA (Multivariate Functional PCA) to encode temporal kinetics into functional principal components.
- **CNN Feature Embedding:** 3D ResNet-18 or DenseNet-40 applied to CT mandible crops at each timepoint. Feature vectors (512-dim) extracted from penultimate layer. Optionally replaced by Vision Transformer (ViT) patch embeddings for better spatial attention.
- **Biomarker Normalization:** Log-transform cytokine values (log-normal distribution). Impute missing timepoints using MissForest (non-parametric). Standardize to z-scores per analyte.
- **Dosimetric Encoding:** Pass DVH vectors through a 2-layer MLP encoder. 3D dose maps: use pre-trained 3D DenseNet-40 to extract dose embedding.

▼

STAGE 3 — TEMPORAL FUSION MODEL

- **Architecture — Temporal Fusion Transformer (TFT):** Encodes all four feature streams across T0–T4 using self-attention over the time axis. Variable selection network identifies which

input branch contributes most at each timepoint. Output: hidden state per patient per timepoint.

- **Alternative — LSTM with Late Fusion:** Separate LSTM per branch (imaging, biomarkers, dosimetric, clinical) across T0–T4. Concatenate hidden states at final timestep → dense layer. Simpler; preferred if n < 100.
- **Competing-Risk Survival Head:** DeepHit or Fine-Gray subdistribution model appended to TFT output. Models time-to-ORN-progression and time-to-death as competing events. Outputs cause-specific hazard function over time.
- **Explainability — SHAP + Attention Maps:** SHAP values on final dense layer to rank feature importance per prediction. TFT's temporal attention weights visualised as heatmaps across timepoints and branches.

▼

STAGE 4 — CLINICAL OUTPUT

- **Output 1 — ORN Progression Risk Score:** Continuous 0–1 score per patient per timepoint. High score → expedite imaging review and specialist referral.
- **Output 2 — Current CT Grade Classification:** Multiclass output (Grade 0–4 per Patil et al. 2026 system). Replaces subjective clinician grading with an automated, reproducible classification.
- **Output 3 — Treatment Triage Recommendation:** 4-class output: (a) Conservative management; (b) PENTOCLO protocol; (c) Hyperbaric oxygen therapy; (d) Surgical resection. Confidence interval displayed per recommendation.
- **Output 4 — SHAP Explanation Card:** Per-patient report showing which features (e.g., 'IL-6 spike at T2', 'mandibular basilar erosion on T3 CT') most influenced the recommendation.

5.2 Pseudocode

The following pseudocode describes the core training and inference logic for the proposed system.

```
# STAGE 1: DATA LOADING
LOAD patient_records  # n ~ 100-150 ORN-confirmed patients from Tata Memorial
FOR each patient p IN patient_records:
    CT_series[p] = load_dicom(p, timepoints=[T0, T1, T2, T3, T4])
    mandible[p]  = nnUNet_segment(CT_series[p])   # auto-segmentation
    dose_map[p]  = load_rt_plan(p)                # 3D dose distribution
    biomarkers[p]= load_cytokines(p, markers=[IL6, IL8, TNFa, MCP1, CTX])
    clinical[p]  = load_clinical(p, fields=[tobacco, diabetes, chemo, stage])

# STAGE 2: FEATURE EXTRACTION
FOR each patient p:
    radiomics[p]  = pyradiomics_extract(mandible[p])  # 1,645 features x 5
timepoints
    fpca_scores[p]= MFPCA(radiomics[p], n_components=10)
    cnn_embed[p]  = ResNet18_forward(mandible[p])     # 512-dim per timepoint
    dvh[p]        = extract_DVH(dose_map[p], roi='mandible')
    dose_embed[p] = MLP_encoder(dvh[p])               # 64-dim embedding
```

```
    bm[p]          = log_transform(biomarkers[p])
    bm[p]          = MissForest_impute(bm[p])
    bm[p]          = z_score_normalise(bm[p])
    X[p] = concat([fpca_scores[p], cnn_embed[p], dose_embed[p], bm[p],
clinical[p]])

# STAGE 3: TEMPORAL FUSION TRANSFORMER (TFT)
MODEL = TemporalFusionTransformer(
    input_dim=F, hidden_dim=128, n_heads=4, n_layers=3, dropout=0.2,
    output_heads=['progression_score', 'grade_class', 'treatment_class']
)

# STAGE 4: TRAINING LOOP
SPLIT data -> train (70%), val (15%), test (15%)  # stratified by grade
LOSS = alpha*BCELoss(progression_score) + beta*CrossEntropy(grade_class)
       + gamma*CrossEntropy(treatment_class) + delta*DeepHit_loss(survival)
FOR epoch IN range(200):
    y_hat = MODEL(X_train)
    loss  = compute_loss(y_hat, y_train)
    loss.backward()
    optimizer.step()   # Adam, lr=1e-4
    IF val_loss no improvement for 20 epochs: BREAK

# STAGE 5: EXPLAINABILITY
shap_values = SHAP_TreeExplainer(MODEL, X_test)
attention   = MODEL.get_attention_weights(X_test)

# STAGE 6: EVALUATION
COMPUTE AUC, F1, Brier_score, calibration_curve(y_test, y_hat)
COMPARE vs PREDMORN NTCP baseline (Humbert-Vidan 2025)
```

5.3 Technology Stack Summary

Component	Technology	Purpose
Segmentation	nnU-Net (PyTorch)	Auto-segment mandible from CT DICOM
Radiomics	PyRadiomics (Python)	Extract 1,645 texture/shape features per CT
Temporal Encoding	MFPCA (scikit-fda)	Encode time-series radiomics into functional PCs
Deep Learning	PyTorch + HuggingFace	3D ResNet/ViT image embeddings; TFT architecture
Survival Modeling	DeepHit / lifelines	Competing-risk time-to-event modeling
Biomarker Imputation	MissForest (R/sklearn)	Non-parametric missing value imputation
Explainability	SHAP (shap library)	Feature attribution per prediction
Experiment Tracking	MLflow / Weights & Biases	Hyperparameter search, reproducibility

Component	Technology	Purpose
Clinical Output UI	`Streamlit (Python)`	Prototype clinician dashboard (optional)

6. Plan of Work — Next Semester

The plan below is structured as a six-month phased roadmap. It is designed to yield a minimum viable first paper (ML-driven ORN grade classification) by Month 4, while laying the infrastructure for the full longitudinal fusion model as the semester-end deliverable.

Phase / Timeline	Milestone	Key Tasks
Phase 1 Months 1-2	**Systematic Review & Data Design**	• Complete systematic review: all ORN ML papers (PREDMORN, Reber, Hosseinian, Barua, West et al.) • Design data collection schema: imaging timepoints, biomarker panel, clinical variables • IRB / ethics application submission at Tata Memorial Hospital • Identify retrospective ORN-confirmed HNC cases from Tata records (2015-2024) • Pilot TCIA public HNC dataset (MD Anderson) for architecture testing
Phase 2 Months 2-3	**Data Collection & Preprocessing**	• DICOM de-identification and CT series extraction for all identified patients • Mandible auto-segmentation using nnU-Net; radiologist QA on 20% of cases • Grade each patient using Patil et al. 2026 CT system (expert radiologist + blinded reviewer) • PyRadiomics feature extraction per timepoint per sub-volume • Biomarker data extraction from Tata biobank or clinical records (if available)
Phase 3 Months 3-4	**Baseline Model Development**	• Train baseline classifiers: XGBoost, Random Forest on radiomics + clinical features • LSTM model on serial CT feature vectors across T0-T4 • Single-timepoint CNN (ResNet-18) on CT crops as ablation baseline • Evaluate: AUC, F1, calibration; compare to PREDMORN NTCP benchmark
Phase 4 Months 4-5	**Temporal Fusion Model**	• Implement Temporal Fusion Transformer (TFT) — PyTorch / pytorch-forecasting library • Integrate all four branches: imaging, dosimetric, biomarker, clinical

Phase / Timeline	Milestone	Key Tasks
		• Competing-risk survival head (DeepHit) for time-to-progression • Hyperparameter tuning via Optuna; tracked with MLflow
Phase 5 Months 5-6	**Validation, Explainability & Write-up**	• SHAP analysis: generate per-patient explanation cards • Temporal attention weight visualisation across branches and timepoints • Compare model output against inter-clinician agreement benchmarks • Draft manuscript — target: Oral Oncology or Radiotherapy & Oncology • Prepare conference abstract for ASTRO / MASCC 2026

7. Risk Assessment & Feasibility Analysis

This section honestly addresses the principal risks of Angle 2 and the mitigation strategies available. The research is ambitious; acknowledging its challenges is a prerequisite to managing them.

Risk	Impact	Mitigation Strategy
Insufficient retrospective ORN patients with serial imaging (target n=100+)	High	Begin case identification immediately via Tata's radiation oncology and oral surgery records (2015-2024). Supplement with prospective enrolment if n < 60. Consider federated collaboration with AIIMS Delhi.
Biomarker data (IL-6, CTX) not available in existing records	High	Downgrade biomarker branch to 'optional module' in Phase 1. Core model uses imaging + clinical + dosimetric only. Biomarker arm becomes a prospective add-on or separate pilot paper.
IRB/ethics approval delays at Tata Memorial	Medium	Submit IRB application in Month 1 in parallel with literature review. Use retrospective waiver pathway for de-identified imaging data. Ethics consultation during Phase 1.
Class imbalance: ORN progression events rare (Grade 0 >> Grade 3+)	Medium	Apply SMOTE oversampling for training; class-weighted loss functions. Use calibrated probability outputs rather than hard classification thresholds.
Mandible segmentation failures on low-quality or old CT scans	Medium	Manual QA on 100% of segmentations in first 50 cases. Fine-tune nnU-Net on Tata's specific CT scanner/protocol characteristics.
TFT model underfits due to small n (n < 100)	Medium	Pre-specify LSTM with late fusion as fallback. Conduct power analysis post-data-collection. Use transfer learning from TCIA HNC dataset for imaging branch.
One semester insufficient for full model completion	Low	Phase the deliverable: Grade classification paper (Phases 1-3) as primary semester output; longitudinal TFT as ongoing research. This is explicitly acknowledged as a 2-3 semester project.

8. Ethical Considerations

8.1 Patient Data Governance

All patient data used in this research will be handled in strict compliance with Indian data protection law (DPDP Act, 2023) and the ICMR National Ethical Guidelines for Biomedical and Health Research Involving Human Participants (2017). The following protections will be implemented:

- **De-identification:** All DICOM imaging data will be de-identified using a validated pipeline prior to analysis. Patient names, dates of birth, and hospital identifiers will be removed; dates will be shifted by a randomized offset preserving temporal intervals.
- **Minimal necessary data:** Only variables directly required by the model will be extracted. Social history (tobacco type, alcohol) will be coded categorically, not verbatim.
- **Secure storage:** All data will be stored on Tata Memorial's institutional servers or a HIPAA-equivalent cloud environment. No patient data will be transferred to personal computing devices.

- **IRB oversight:** Ethics committee approval will be sought from Tata Memorial Hospital's Institutional Review Board before any data extraction. A retrospective waiver of consent will be requested for de-identified historical data.

8.2 Model Fairness and Bias

The South Asian cohort, while a scientific strength, also introduces bias considerations that must be explicitly managed:

- **Treatment access bias:** Hyperbaric oxygen therapy is not uniformly available across India. Treatment triage outputs must be validated against actual treatment received, with documentation of access constraints that may have influenced historical treatment decisions.
- **Socioeconomic confounding:** Tobacco type (bidi vs. cigarette vs. smokeless) is correlated with socioeconomic status at Tata Memorial. The model must include stratified analysis to ensure it does not proxy socioeconomic deprivation as a clinical risk factor.
- **Generalizability disclosure:** Any publication will explicitly state that the model was trained on a South Asian cohort and may require recalibration before deployment in Western populations. The model is not positioned as universally generalizable.

8.3 Clinical Safety

The treatment triage output (Output 3) carries the highest clinical risk of any model component. The following safeguards are mandatory:

- **No autonomous recommendation:** The model will be positioned as a decision-support tool, not a replacement for clinical judgment. All outputs will include confidence intervals and SHAP explanation cards.
- **Validation against clinician benchmark:** Model recommendations will be compared against the MD Anderson interobserver agreement benchmark (2025). The model will not be promoted to clinical use unless it demonstrably exceeds median clinician accuracy.
- **Surgeon review gate:** In the proposed prototype dashboard, the surgical resection recommendation (the highest-intensity intervention) will require an additional confirmation step and cannot be triggered by model output alone.

9. Statement of Novelty

The following table summarises how this research advances the field beyond every existing publication, across five dimensions of novelty.

Dimension	Current State-of-the-Art	This Research
Temporal modeling	All models: single pre-treatment timepoint snapshot	Serial data across 5 timepoints (T0-T4); TFT over time axis
Modality integration	Imaging only OR clinical only OR dosimetric only	All four simultaneously: imaging + biomarkers + dosimetric + clinical
Clinical output	Binary ORN yes/no, or grade at single timepoint	Progression risk score + grade + treatment triage + SHAP card

Dimension	Current State-of-the-Art	This Research
Population	MD Anderson (US) or European institutions exclusively	South Asian cohort: bidi/smokeless tobacco, oral cavity primaries
Survival modeling	Not applied to ORN ML literature	Competing-risk DeepHit: time-to-progression vs. time-to-death

WHY THIS HAS NOT BEEN DONE BEFORE

The confluence of three factors has prevented this research from being conducted previously: (1) **data infrastructure** — longitudinal, multi-modal ORN datasets require years of structured follow-up and are not available from existing public repositories; (2) **institutional access** — Tata Memorial's unique combination of high ORN case volume, established CT grading taxonomy (Patil et al., 2026), and South Asian demographics creates a data environment that cannot be replicated outside India; and (3) **algorithmic readiness** — the Temporal Fusion Transformer architecture was only published in 2021 and has not yet been applied to oncological complication monitoring.

References

1. Humbert-Vidan L et al. Comparison of ML methods for prediction of ORN in HNC. *Br J Radiol.* 2021;94:20200026. DOI: 10.1259/bjr.20200026

2. Reber B et al. Comparison of ML and DL for prediction of ORN. *Adv Radiat Oncol.* 2023;8(4):101163. DOI: 10.1016/j.adro.2022.101163

3. Humbert-Vidan L et al. PREDMORN multi-institutional NTCP model. *Int J Radiat Oncol Biol Phys.* 2025. PMC article

4. Hosseinian S et al. Cluster-based toxicity estimation of ORN via unsupervised ML. *medRxiv.* 2023. DOI: 10.1101/2023.03.24.23287710

5. West NA et al. Clinical and dosimetric dataset for longitudinal ORNJ NTCP. *Sci Data.* 2026;13:16. DOI: 10.1038/s41597-025-06321-w

6. Patil V et al. Decoding ORN: Radiological Progression and Novel CT-Based Grading System. *Cancers.* 2026;18:187. DOI: 10.3390/cancers18020187

7. Principe S, Dikova V, Bagan J. Salivary Cytokines in HNC patients treated with RT. *J Clin Exp Dent.* 2019;11(11):e1072-7. DOI: 10.4317/jced.56318

8. Principe S et al. Salivary IL-8 as predictive biomarker of RT response. *Clin Oral Investig.* 2022;26:437-448. DOI: 10.1007/s00784-021-04017-0

9. Watson EE et al. ClinRad ORN classification system. *J Clin Oncol.* 2024;42(16):1922-1933. DOI: 10.1200/JCO.23.01951

10. Singh A et al. Osteoradionecrosis of the jaw: A mini review. *Front Oral Health.* 2022;3:980786. DOI: 10.3389/froh.2022.980786

11. Lim JW et al. Temporal Fusion Transformers for interpretable multi-horizon time series forecasting. *Int J Forecast.* 2021;37(4):1748-1764. DOI: 10.1016/j.ijforecast.2021.03.012

12. Lee C, Yoon J, van der Schaar M. Dynamic-DeepHit: A deep learning approach for dynamic survival analysis. *IEEE Trans Biomed Eng.* 2020;67(12):3351-3361. DOI: 10.1109/TBME.2020.2980091